Physics-Closed Multimodal Dense Prediction via Latent Continuum Equivalency: Factor-of-Safety Cells and Mechanistic Path Equilibrium Fusion

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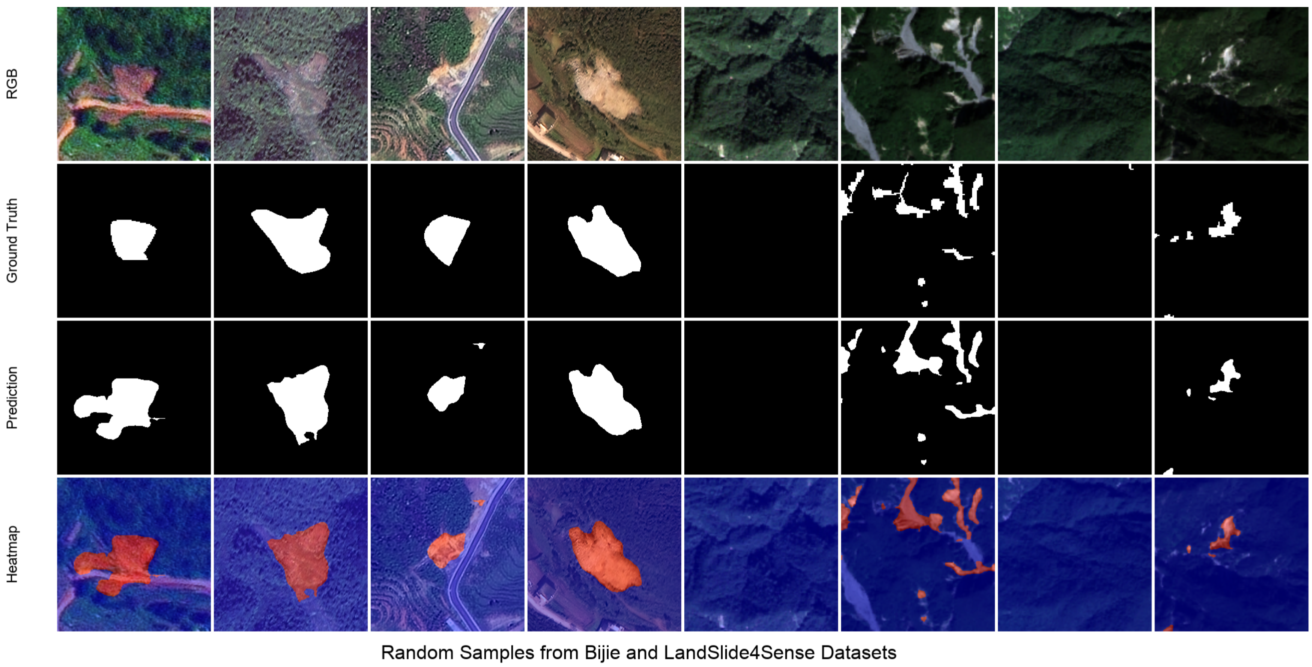


Figure 1: Opening illustration of dense multimodal segmentation under anisotropic optical-topographic fusion (RGB, ground truth, prediction, confidence heatmap). Landslide inventory mapping is used as a stress test of the proposed physics-closed continuum operators.

Abstract

Multimodal dense predictors often fuse heterogeneous streams with unconstrained statistical gates, even when one modality encodes a continuum mechanical field rather than an appearance cue. We ask whether a single reduced-order continuum operator can be injected into high-dimensional networks without destabilizing backpropagation, and whether that operator can serve as a commensurable currency for both feature gating and dual-path fusion. We introduce **Latent Continuum Equivalency (LCE)**: the same Taylor-stabilized infinite-slope factor-of-safety (FS) skeleton appears at the encoder inlet, hybrid bridges, skip lattice, dual physics decoders, and output routing. Material parameters use bounded exponentials $c = \exp(\mathrm{clip}(w))$, which we prove place resisting/driving forces on a compact set and yield a locally Lipschitz FS gate. Dual-path logits are merged by **Mechanistic Path Equilibrium Fusion (MPEF)**, the softmax of failure energies, which we identify as the unique maximizer of an entropy-regularized linear energy on the simplex. Instantiated as GeoPhysicsLandslideNet (PS-GPLNet), a pentastream CNN-physics-foundation architecture, the method is evaluated on Bijie and Landslide4Sense, anisotropic topographic-spectral segmentation benchmarks with severe imbalance. PS-GPLNet attains best-epoch F1/IoU of $0.933/0.907$ on Bijie and $0.736/0.660$ on Landslide4Sense, exceeding strong CNN, transformer, and dual-stream baselines under a unified protocol. Empirical GPU probes confirm material compactness, local FS-gate sensitivity bounds, and the MPEF energy identity. Inference studies further show structured fuse3 latents (t-SNE), Boundary F-score/Hausdorff fidelity, and flatter F1 decay under RGB+DEM Gaussian noise than U-Net and DeepLabV3+.

1 Introduction

Dense prediction from multiple sensors is a core problem in computer vision [1, 2, 3]. When modalities are correlated, as in RGB-D, learned fusion can exploit statistical redundancy. When modalities are *anisotropic*, one stream may encode an appearance field while another encodes a physical continuum (slope, shear, height), and unconstrained gates are free to ignore mechanics that the task actually depends on [4, 5].

A growing line of physics-informed learning places continuum or geometric constraints in losses or attention [6, 7, 4, 8, 5, 9]. Most susceptibility-oriented geo-AI work still keeps physics at the objective [10, 11], while dual-stream landslide segmenters keep fusion statistical [12, 13]. We instead close a reduced-order continuum loop *inside* the forward pass.

Our contribution is a general operator family for multimodal dense prediction. We formalize Latent Continuum Equivalency (LCE): every stage applies an operator from the same infinite-slope FS class, so failure energy remains a shared physical currency from inlet to outlet. Bounded material reparametrization keeps FS finite and the sigmoid gate locally Lipschitz. MPEF routes dual decode paths by softmax failure energy and is the unique entropy-regularized energy maximizer on the simplex [14]. Landslide segmentation on Bijie [15] and Landslide4Sense [16] is the stress test: spectral ambiguity plus topography governed by gravity/shear, not RGB-D correlation. Under a unified protocol, PS-GPLNet reaches F1/IoU 0.933/0.907 (Bijie) and 0.736/0.660 (L4S).

2 Related Work

Dense segmentation and multimodal fusion. U-Net [1], DeepLabV3+ [2], SegFormer [3], and TransUNet [17] dominate dense masks. Multimodal optical-DEM fusion improves landslide mapping [15, 16, 13, 18]. Dual-stream gated U-Nets [12] learn scalar gates; we use them only as empirical baselines. Interpretable fusion via unfolded sparse coding [9] motivates treating fusion as an optimization identity rather than a free MLP.

Geometric and topological priors in vision. Virtual Normal enforces high-order 3D geometry beyond pixel errors [4]. Topological PH losses inject discrete structure with differentiability arguments [8]. Lipschitz certificates for segmentation quantify robustness [19, 20]. Our FS gate plays an analogous role for continuum stability.

Physics inside architectures. Gravityformer multiplies transformer attention by a closed-form gravity interaction [5]. PINNs place PDE residuals in losses [6, 7]. Landslide PINNs largely target susceptibility [10, 11, 21]. PS-GPLNet differs by architectural closure of FS cells across encoding, fusion, decoding, and path equilibrium, with a foundation stream (Prithvi+LoRA [22, 23]).

Table 1: Conceptual positioning for dense multimodal prediction.

Approach	Dense mask	Continuum in forward	Foundation
CNN / ViT segmenters	✓	–	optional
Dual-stream gated [12]	✓	–	rare
Susceptibility PINN / PGML	often raster	loss / head	rare
Gravity-/geometry-informed CV [5, 4]	task-dep.	attention / loss	rare
PS-GPLNet (ours)	✓	FS cells + MPEF	Prithvi+LoRA

Positioning. Table 1 contrasts dense masks, FS-in-forward, and foundation context. The novelty claim is LCE + MPEF, not a new remote-sensing dataset.

3 Method

We present a physics-closed multimodal dense predictor whose forward pass is organized by a single reduced-order continuum narrative: the infinite-slope factor of safety (FS). The classical geotechnical FS is not wired verbatim into neurons. We first recall the textbook form, then derive a Taylor-stabilized neural operator $\mathrm{FS}_{\mathrm{nn}}$ that relocates pore pressure into an additive driving term and admits a latent variant without trigonometric angles. That operator family is then realized end-to-end in GeoPhysicsLandslideNet (PS-GPLNet) as four stages. Stage A (inlet) comprises five parallel encoders together with physics proxies and pixel/latent mechanistic cells. Stage B (hybridization) applies Complementary Modality Bridges (CMB) that resonate CNN texture with physics pyramids. Stage C (tri-stream fusion) places MAO-GeoEGCA at the bottleneck and TTEB multi-scale skips. Stage D (dual decode + equilibrium) runs dual PhysicsDecoders with PGDI, then MPEF instability routing at three supervision heads. The remainder of this section follows that order after the FS derivation.

3.1 Problem and notation

Let $x_1 \in \mathbb{R}^{B\times 3\times H\times W}$ denote the optical stream (RGB) and $x_2 \in \mathbb{R}^{B\times 3\times H\times W}$ the topographic/auxiliary stream. On Landslide4Sense, x_2 stacks NDVI, slope, and DEM; on Bijie, DEM is replicated across three channels. The prediction target is a binary landslide mask $y \in \{0,1\}^{B\times 1\times H\times W}$. Unless stated otherwise we use $H = W = 256$ at the network inlet and a unified fusion width $C = 64$.

From (x_1, x_2) the network extracts normalized physics proxies (s, d, v) (slope, elevation, and vegetation) and maps them to geotechnical variables (α, h, m), where α is the slope angle in radians (pixel-wise), $h > 0$ is a soil-column / elevation height proxy, and $m \in (0, 1)$ is a moisture-saturation proxy. Separate RGB-path and DEM-path mappers produce stream-specific triples (α_r, h_r, m_r) and (α_d, h_d, m_d). We write $\odot$ for Hadamard product, $[\cdot \| \cdot]$ for channel concatenation, and $\varepsilon > 0$ for a numerical floor (typically 10^{-6}).

3.2 Classical infinite-slope factor of safety

For an infinite slope with seepage parallel to the surface, the textbook factor of safety compares resisting and driving forces on a potential slip plane of depth H [24]:

$$\begin{aligned} \mathrm{FS}_{\text{class}} &= \frac{c' + \big(\gamma_{\text{sat}} H - u\big)\cos^2\alpha\,\tan\phi'}{\gamma_{\text{sat}} H \sin\alpha\cos\alpha}, \\ u &= m\,\gamma_w H\cos^2\alpha. \end{aligned} \tag{1}$$

Here c' is effective cohesion, ϕ' the effective friction angle, γ_{sat} the saturated unit weight, γ_w the unit weight of water, α the slope angle, and $m \in [0,1]$ the saturation ratio. Substituting u yields

$$\mathrm{FS}_{\text{class}} = \frac{c' + \big(\gamma_{\text{sat}} - m\gamma_w\big) H\cos^2\alpha\,\tan\phi'}{\gamma_{\text{sat}} H\sin\alpha\cos\alpha}. \tag{2}$$

Failure is declared when $\mathrm{FS}_{\text{class}} \leq 1$. Equation (2) is a valid engineering calculator for hand analysis, but it is a poor neuron. The term $-m\gamma_w$ subtracts from the resisting numerator: early in training, uncalibrated material weights can drive that numerator through zero or negative values, flipping gradient signs and producing NaNs. Moreover, embedding $\sin\alpha$ and $\cos\alpha$ into deep latent tensors, where channels no longer represent geodetic angles, is physically meaningless and numerically brittle. We therefore derive a neural form that (i) relocates hydrology into an additive driving term and (ii) admits a latent counterpart that drops angles entirely while preserving the resisting/driving ratio skeleton.

3.3 Derivation of the Taylor-stabilized neural FS

A dense predictor scores *instability*. Define stress as the reciprocal $S = 1/\mathrm{FS}_{\text{class}}$. Expanding (2),

$$S = \frac{A}{B - C}, \tag{3}$$

with the grouped forces

$$\begin{aligned} A &= \gamma_{\text{sat}} H\sin\alpha\cos\alpha, \\ B &= c' + \gamma_{\text{sat}} H\cos^2\alpha\,\tan\phi', \\ C &= m\gamma_w H\cos^2\alpha\,\tan\phi'. \end{aligned} \tag{4}$$

Here A is gravity drive, B dry structural resistance, and C hydrological destablization. Whenever $C \geq B$, the denominator of (3) vanishes or turns negative, which is catastrophic for backpropagation.

Proposition 1 (First-order Taylor relocation of pore pressure)**.** *Assume $C/B < 1$ in a neighbourhood of dry soil ($C = 0$). Then*

$$\begin{aligned} \frac{A}{B-C} &= \frac{A}{B}\,\frac{1}{1 - C/B} \\ &= \frac{A}{B}\sum_{k=0}^{\infty}\Big(\frac{C}{B}\Big)^k \\ &= \frac{A}{B} + \frac{AC}{B^2} + O\big((C/B)^2\big). \end{aligned} \tag{5}$$

Truncating after the linear term yields the first-order structural equivalence

$$\frac{A}{B-C} \;\propto\; \frac{A+C}{B} \tag{6}$$

up to a positive local scale B^{-1}. Consequently, subtracting hydrology from resistance is, to first order, equivalent to adding *hydrology into the driving reference frame.*

Proof. Factor B from the denominator of (3) and apply the geometric series $1/(1-x) = \sum_{k\geq 0} x^k$ for $|x| < 1$ with $x = C/B$. Truncation after $k = 1$ gives (5). Multiplying numerator and denominator of the truncated form by B rearranges to the proportional additive drive (6). $\square$

Physically, the classical layout treats water as a lubricant that reduces effective normal stress; the neural layout treats water as an additive downslope driver. Both describe the same tipping point (driving forces overtaking resistance), but the neural layout keeps all tensors strictly positive. Reverting from stress S to an FS-shaped ratio and absorbing material constants into learnable positive weights (c, ϕ, γ, w_m) produces the operator used in pixel cells and in MPEF.

Definition 1 (Taylor-stabilized neural FS)**.**

$$\begin{aligned} &\mathrm{FS}_{\text{nn}}(\alpha, h, m) \\ &\quad= \frac{c + \phi\, h\cos^2\alpha}{\gamma\, h\sin\alpha\cos\alpha + w_m m + \varepsilon}. \end{aligned} \tag{7}$$

Failure energy and the pixel gate are

$$\begin{aligned} \mathrm{FE} &= \mathrm{ReLU}\big(1 - \mathrm{FS}_{\text{nn}}\big), \\ g &= \sigma\big(\psi - \mathrm{FS}_{\text{nn}}\big). \end{aligned} \tag{8}$$

Equation (7) is deliberately not identical to (1): pore pressure has been moved from a subtractive numerator into an additive denominator by Proposition 1, and $(\tan\phi', \gamma_{\text{sat}}, \gamma_w)$ have been reparameterized into bounded learnable positives (Definition 2). This is the form that replaces generic affine neurons in mechanistic cells.

Definition 2 (Bounded material reparametrization)**.** *For each material logit $w \in \{w_c, w_\phi, w_\gamma, w_m\}$,*

$$\text{positive\_scale}(w) = \exp\big(\mathrm{clip}(w; -8, 8)\big), \tag{9}$$

so $c, \phi, \gamma, w_m \in [e^{-8}, e^{8}]$. Clipping before the exponential prevents overflow while preserving positivity and smooth gradients almost everywhere.

3.4 Pixel and latent mechanistic cells

PixelMechanisticCell. At (or near) full spatial resolution, the DEM-derived angle α is a known geometric constant per pixel. Hardcoding $\sin\alpha\cos\alpha$ and $\cos^2\alpha$ into the gate is therefore legitimate: the network need not rediscover the trigonometric landscape of gravity. The cell applies a 1×1 convolution and multiplies by the FS gate:

Definition 3 (PixelMechanisticCell)**.**

$$
\begin{aligned}
&\mathrm{PixelMechanisticCell}(x, \alpha, h, m) \\
&\quad = \mathrm{Conv}_{1\times 1}(x) \\
&\qquad \odot\ \sigma\big(\psi - \mathrm{FS}_{\mathrm{nn}}(\alpha, h, m)\big).
\end{aligned} \tag{10}
$$

When (α, h, m) and x differ in spatial size, the proxies are bilinearly resized to match x. If mechanistic gating is disabled (ablation), the cell returns the convolution alone. Conceptually, low $\mathrm{FS}_{\mathrm{nn}}$ (unstable slopes) opens the gate and admits features; high $\mathrm{FS}_{\mathrm{nn}}$ (stable slopes) attenuates them.

LatentMechanisticCell. Deeper feature maps do not live in a geodetic coordinate frame. Applying $\sin\alpha$ or $\cos\alpha$ to abstract channels would force angular geometry onto tensors that no longer represent angles, exactly the pathology we seek to avoid. We therefore drop trigonometry and realize the same resisting/driving *ratio skeleton* with softplus projections:

Definition 4 (LatentMechanisticCell)**.**

$$
\begin{aligned}
R &= \mathrm{Softplus}(W_R * H), \\
D &= \mathrm{Softplus}(W_D * H),
\end{aligned} \tag{11}
$$

$$
\begin{aligned}
&\mathrm{LatentMechanisticCell}(H) \\
&\quad = \mathrm{Conv}_{3\times 3}\Big(\mathrm{LeakyReLU}\big(\psi - \tfrac{R}{D+\varepsilon}\big)\Big).
\end{aligned} \tag{12}
$$

Thus $\mathrm{FS}_{\mathrm{nn}}$ and the latent ratio $R/(D+\varepsilon)$ belong to one continuum class: pixel cells keep DEM angles; latent cells keep only the continuum ratio. This dual realization is the operational meaning of Latent Continuum Equivalency at the cell level.

PhysicsEncoder. Each PhysicsEncoder stacks a pixel stem followed by successive latent cells with downsampling, then projects every level to width C:

$$
\begin{aligned}
f_0 &= \mathrm{Stem}(\mathrm{PixelMech}(x, \alpha, h, m)), \\
f_{\ell+1} &= \mathrm{LatentMech}(\mathrm{Down}(f_\ell)), \quad \ell = 0, \dots, 3, \\
P^L &= \mathrm{Proj}_L(f_L), \quad L = 0, \dots, 4.
\end{aligned} \tag{13}
$$

At 256×256 input the physics pyramid lives on $\{256, 128, 64, 32, 16\}$. Two independent PhysicsEncoders run on RGB and DEM with their own proxies, yielding $\{P^L_{\mathrm{rgb}}\}$ and $\{P^L_{\mathrm{dem}}\}$.

3.5 Physics proxies and foundation inlet

Proxy extraction. Dataset channels are min-max normalized to $[0, 1]$. On Landslide4Sense, $s = x_2[:, 1]$, $d = x_2[:, 2]$, $v = x_2[:, 0]$. On Bijie (replicated DEM),

$$
\begin{aligned}
d &= x_2[:, 0], \\
s &= \mathrm{Norm}(\|\nabla d\|), \\
v &= \mathrm{Norm}\Big(\tfrac{G-R}{G+R+\varepsilon}\Big),
\end{aligned} \tag{14}
$$

where ∇d is a Sobel gradient and G, R are green/red bands of x_1.

PhysicsProxyMapper. Separate RGB-path and DEM-path mappers (identical architecture, independent weights) produce

$$
\begin{aligned}
\alpha &= \tfrac{\pi}{2}\,\sigma(w_\alpha s + b_\alpha), \\
h &= \mathrm{Softplus}(w_h d + b_h) + \varepsilon, \\
m &= \sigma(w_m v + b_m).
\end{aligned} \tag{15}
$$

Softplus keeps $h > 0$; sigmoid keeps $m \in (0, 1)$; the $\pi/2$ scale maps the slope logit into a valid angle range.

Foundation stack. A six-band observed stack feeds Prithvi-EO with LoRA [22, 23]: on Landslide4Sense, $[R, G, B, \mathrm{NDVI}, \mathrm{slope}, \mathrm{DEM}]$; on Bijie, $[R, G, B, \mathrm{DEM}, \mathrm{Sobel}(d), \mathrm{vegIndex}]$. Intermediate transformer blocks are projected to width C and resized onto a five-level pyramid $\{T^L_{\mathrm{fm}}\}$.

3.6 Stage A-B: Pentastream encoding and Complementary Modality Bridge

Stage A runs five encoders in parallel: E_{rgb} (EfficientNet [25] on RGB with pyramid $\{E^i_{\mathrm{rgb}}\}_{i=0}^4$), P_{rgb} (PhysicsEncoder on RGB with (α_r, h_r, m_r)), E_{dem} (EfficientNet on DEM with pyramid $\{E^i_{\mathrm{dem}}\}$), P_{dem} (PhysicsEncoder on DEM with (α_d, h_d, m_d)), and T_{fm} (Prithvi+LoRA foundation pyramid). EfficientNet-B4 at 256×256 yields spatial grids $\{128, 64, 32, 16, 8\}$ with growing channel counts. Physics and Prithvi pyramids use different native grids; they are bilinearly aligned to each EfficientNet level via a fixed legacy index map $\mathrm{LEGACY} = (1, 2, 3, 4, 4)$, so EffNet index i consumes physics level $\mathrm{LEGACY}[i]$. CNN texture is *complemented*, not replaced: CMB increases trust in physics only when CNN and physics co-activate (resonate).

Definition 5 (Complementary Modality Bridge)**.** *Let* $\tilde{E} = \mathrm{GN}(\mathrm{ReLU}(\mathrm{Conv}_{1\times 1}(E)))$ *project the CNN tensor to width* C *and* $\tilde{P} = \mathrm{GN}(P)$ *normalize the aligned physics tensor. Then*

$$
g = \sigma\big(\mathrm{Conv}_{1\times 1}(\tilde{E} \odot \tilde{P})\big), \tag{16}
$$

$$
B_{\mathrm{mix}} = \mathrm{Conv}_{1\times 1}([\tilde{E} \,\|\, \tilde{P}]), \tag{17}
$$

$$
H = g \odot \tilde{P} + (1 - g) \odot \tilde{E} + B_{\mathrm{mix}}. \tag{18}
$$

The gate g opens toward physics when element-wise agreement $\tilde{E} \odot \tilde{P}$ is strong; otherwise appearance features dominate. The residual mix B_{mix} preserves a shared channel pathway so neither stream is hard-gated to zero. Applying CMB at every EffNet index produces hybrid pyramids $\{H^i_{\mathrm{rgb}}\}$ and $\{H^i_{\mathrm{dem}}\}$. Foundation maps are likewise aligned to E^i_{rgb}, yielding $\{\hat{T}^i_{\mathrm{fm}}\}$. Subsequent fusion consumes only these hybrids and foundation maps, not raw EffNet features. Algorithm 1 states the exact operator used in code.

3.7 Stage C: MAO-GeoEGCA (bottleneck fusion)

Stage C cross-references local geomechanics with planetary-scale context. At neck and bottleneck levels $i \in \{2, 3\}$ (spatial

Require: `CNN features` E, `physics` P

$P \leftarrow \text{match\_spatial}(P, E)$
$\tilde{E} \leftarrow \text{GN}(\text{ReLU}(\text{Conv}_{1\times1}(E)))$
$\tilde{P} \leftarrow \text{GN}(P)$
$g \leftarrow \sigma(\text{Conv}_{1\times1}(\tilde{E} \odot \tilde{P}))$
$B \leftarrow \text{Conv}_{1\times1}([\tilde{E}; \tilde{P}])$
return $g \odot \tilde{P} + (1-g) \odot \tilde{E} + B$

Algorithm 1: Complementary Modality Bridge (CMB)

grids 32×32 and 16×16 for B4@256), Manifold-Aligned Orthogonal Geo-EGCA (**MAO** = manifold-aligned orthogonal; **GeoEGCA** = Geo-Equilibrium Gated Cross-Attention) treats the concatenated RGB/DEM hybrids as a *physics anchor* and Prithvi as environmental context. Restricting MAO to these two levels leaves four $\times2$ upsampling stages in the decoder, restoring full 256×256 logits; applying MAO at 8×8 would under-resolve the decode chain.

Physical Equilibrium Gate. The gate G is a *pixel-wise* spatial mask at the bottleneck resolution (shape $B \times 1 \times H \times W$), not a single patch scalar. After Prithvi tokens are resized to standard 2D maps, G is formed by element-wise resonance $\hat{T} \odot X_p$, a 3×3 depthwise-separable neighbourhood filter, and a sigmoid. Low G suppresses planetary context where local physics indicate equilibrium; high G admits foundation context where instability cues are strong. The Value tensors V carry the foundation *content* that attention weights redistribute; without V, attention would score relevance but transport no payload.

Definition 6 (MAO-GeoEGCA)**.** *Form the physics anchor and equilibrium gate*

$$\begin{aligned} X_p &= \text{Conv}_{1\times1}([\hat{H}_{\text{rgb}} \,\|\, \hat{H}_{\text{dem}}]), \\ G &= \sigma\big(\text{DWSepConv}_{3\times3}(\hat{T} \odot X_p)\big). \end{aligned} \tag{19}$$

Project queries from the anchor and keys/values from the foundation stream,

$$Q = W_q X_p, \quad K = W_k \hat{T}, \quad V = W_v \hat{T}. \tag{20}$$

Manifold alignment normalizes queries and modulates keys,

$$\hat{Q} = \frac{Q}{\|Q\|_2 + \varepsilon}, \qquad K' = K \odot \hat{Q}, \tag{21}$$

so attention is biased toward directions consistent with the physics anchor. Multi-head attention (four heads, $d_h = C/4$) and gated residual readout give

$$\text{Attn} = \text{softmax}\Big(\frac{QK'^{\top}}{\sqrt{d_h}}\Big)V, \tag{22}$$

$$F = \text{Conv}_{3\times3}\big(\text{reshape}(\text{Attn}) \odot G\big) + X_p. \tag{23}$$

When local geomechanics indicate low stress, $G \to 0$ and foundation context is suppressed; when instability cues are strong, $G \to 1$ and Prithvi tokens guide the fused map. Outputs F^2 (neck, $B\times64\times32\times32$) and F^3 (bottleneck, $B\times64\times16\times16$) seed the dual decoder. Algorithm 2 matches the implemented fusion kernel.

Require: $\hat{T}, \hat{H}_{\text{rgb}}, \hat{H}_{\text{dem}}$

$X_p \leftarrow \text{Conv}_{1\times1}([\hat{H}_{\text{rgb}}; \hat{H}_{\text{dem}}])$
$G \leftarrow \sigma(\text{DWSepConv}_{3\times3}(\hat{T} \odot X_p))$
$Q \leftarrow W_q X_p$; $K \leftarrow W_k \hat{T}$; $V \leftarrow W_v \hat{T}$
$\hat{Q} \leftarrow \text{normalize}(Q)$; $K' \leftarrow K \odot \hat{Q}$
$\text{Ctx} \leftarrow \text{MHAttn}(Q, K', V)$
return $\text{Conv}_{3\times3}(\text{reshape}(\text{Ctx}) \odot G) + X_p$

Algorithm 2: MAO-GeoEGCA (neck / bottleneck)

3.8 Stage C (continued): TTEB skip lattice

For each fusion level $L \in \{0, 1, 2, 3\}$, the Tri-Temporal Tri-Stream Bridge builds a skip tensor from a lattice over RGB, DEM, and foundation streams at neighbouring scales $\{L-1, L, L+1\}$ (boundary-clamped). Levels are resized to the present spatial size. The present-phase tokens are mixed as

$$A = \text{Conv}_{1\times1}([\text{pres}_{\text{rgb}} \,\|\, \text{pres}_{\text{dem}} \,\|\, \text{pres}_{\text{fm}}]). \tag{24}$$

Eight off-diagonal lattice nodes (previous/next across streams and cross terms) are concatenated and mixed into a context tensor H_{ctx}. A latent FS residual, the same ratio skeleton as Definition 4, defines a stability map

$$\delta = \left|\psi - \frac{\text{Softplus}(W_R A)}{\text{Softplus}(W_D A) + \varepsilon}\right|. \tag{25}$$

Chunked spatial attention uses queries from A and keys/values from H_{ctx}, with an optional temporal router $\omega = \text{softmax}(W_\tau \delta)$ that emphasizes the present phase. The skip is

$$S^L = A + \text{Conv}_{3\times3}\big(\text{AttnOut} \odot \sigma(\delta)\big). \tag{26}$$

Thus TTEB does not invent a new inductive bias: its stability map δ is LCE applied to multi-scale, multi-stream skips. The resulting list $\{S^0, S^1, S^2, S^3\}$ lives on $\{128^2, 64^2, 32^2, 16^2\}$. Algorithm 3 states the per-level skip construction.

Require: `hybrid/FM pyramids; level` L

`gather` $\{L-1, L, L+1\}$ `per stream; align spatially`
$A \leftarrow \text{Conv}_{1\times1}([\text{pres}_{\text{rgb}}; \text{pres}_{\text{dem}}; \text{pres}_{\text{fm}}])$
$H_{\text{ctx}} \leftarrow$ `mix(8 lattice nodes)`
$\delta \leftarrow |\psi - R(A)/(D(A) + \varepsilon)|$
$\text{AttnOut} \leftarrow \text{SpatialAttn}(A, H_{\text{ctx}})$; `scale by` $\omega(\delta)$
return $A + \text{Conv}_{3\times3}(\text{AttnOut} \odot \sigma(\delta))$

Algorithm 3: TTEB at level L

3.9 Stage D: Dual Physics Decoder, PGDI, and MPEF

Stage D deliberately *re-splits* the fused Stage C knowledge into two competitive decode paths so spectral and topographic evidence can double-check the mask under commensurable continuum operators.

Shared vs. path-specific inputs. Both paths share $\{F^3, F^2, S^L\}$ from MAO/TTEB. Path A injects EfficientNet RGB residual $a_5 = \text{Proj}(E^4_{\text{rgb}})$ with proxies (α_r, h_r, m_r); Path B injects DEM residual b_5 with (α_d, h_d, m_d). The paths do not share hidden states during upsampling; they cooperate only at MPEF.

Dual paths. Both paths apply the same decode grammar (latent mechanistic cells, PGDI, and a final pixel mechanistic head), so evidence is decoded under one continuum class rather than a single shared statistical decoder.

PGDI. At each upsample stage, Physics-Gated Decoder Injection injects a TTEB skip through a learned gate on a latent mechanistic cell:

Definition 7 (PGDI)**.** *With decoder state D and aligned skip $S' = \mathrm{align}(S, D)$,*

$$\begin{aligned} \beta &= \sigma\big(\mathrm{Conv}_{1\times1}([D \,\|\, S'])\big),\\ D' &= D + \beta \odot \mathrm{LatentMechanisticCell}(S'). \end{aligned} \tag{27}$$

The gate β is content-dependent: skips that disagree with the current decode state are attenuated; skips that carry complementary continuum structure are admitted. Concretely, the decode chain widens $16^2 \to 32^2 \to 64^2 \to 128^2 \to 256^2$. At 32^2, the upsampled state is first refined by a Latent Mechanistic Cell and then residual-injected with the MAO neck F^2; immediately afterward, PGDI admits the aligned TTEB skip. After four $\times 2$ stages and a pixel FS head, each path emits deeply supervised logits $(O_{\mathrm{main}}, O_{\mathrm{aux2}}, O_{\mathrm{aux3}})$ of shapes $B{\times}1{\times}256{\times}256$, $B{\times}1{\times}64{\times}64$, and $B{\times}1{\times}32{\times}32$. Algorithm 4 states the skip injection.

Require: decoder state D, skip S
$S' \leftarrow \mathrm{match\_spatial}(S, D)$
$\beta \leftarrow \sigma(\mathrm{Conv}_{1\times1}([D; S']))$
return $D + \beta \odot \mathrm{LatentMechanisticCell}(S')$

Algorithm 4: Physics-Gated Decoder Injection (PGDI)

MPEF. Dual-path logits are merged by Mechanistic Path Equilibrium Fusion, i.e., softmax failure energy of the same $\mathrm{FS}_{\mathrm{nn}}$ derived in Sec. 3.3, not a DiGATe-style scalar GateFuse. Failure energy is computed from image-derived physics proxies (α, h, m), bilinearly resized to each head resolution; it does not replace the decoder hidden state, but supplies the routing currency that adjudicates between candidate masks O_A and O_B:

Definition 8 (MPEF)**.** *For path logits O_A, O_B and proxies (α_p, h_p, m_p),*

$$\mathrm{FE}_p = \mathrm{ReLU}\big(1 - \mathrm{FS}_{\mathrm{nn}}(\alpha_p, h_p, m_p)\big), \tag{28}$$

$$[w_A, w_B] = \mathrm{softmax}([\mathrm{FE}_A, \mathrm{FE}_B]), \tag{29}$$

$$\begin{aligned} O = {}& w_A O_A + w_B O_B \\ &+ \mathrm{Conv}_{1\times1}([O_A \,\|\, O_B]), \end{aligned} \tag{30}$$

$$\begin{aligned} \mathcal{R}_{\mathrm{MPEF}} = -\mathbb{E}\big[& w_A \log(w_A + \varepsilon) \\ &+ w_B \log(w_B + \varepsilon)\big]. \end{aligned} \tag{31}$$

Independent MPEF modules fuse main, aux2, and aux3 heads (three masks during training; inference uses the main head). The entropy regularizer discourages indecisive averaging while preserving the continuum interpretation of w. Algorithm 5 states the merge; Algorithm 6 summarizes the full forward pass.

Require: O_A, O_B; path proxies (α_p, h_p, m_p)
align proxies to logit resolution
$\mathrm{FE}_p \leftarrow \mathrm{ReLU}(1 - \mathrm{FS}_{\mathrm{nn}}(\alpha_p, h_p, m_p))$
$[w_A, w_B] \leftarrow \mathrm{softmax}([\mathrm{FE}_A, \mathrm{FE}_B])$
$O \leftarrow w_A O_A + w_B O_B + \mathrm{Conv}_{1\times1}([O_A; O_B])$
$R \leftarrow \mathrm{mean}(-(w \log(w + \varepsilon)))$
return O, R

Algorithm 5: Mechanistic Path Equilibrium Fusion (MPEF)

Require: streams x_1 (RGB), x_2 (topo/aux)
$(s, d, v) \leftarrow \mathrm{physics\_proxies}(x_1, x_2)$; map (α, h, m)
encode $E_{\mathrm{rgb}}, E_{\mathrm{dem}}, P_{\mathrm{rgb}}, P_{\mathrm{dem}}, T_{\mathrm{fm}}$
for $i = 0..4$: CMB hybrids H^i; align $\hat{T}^i$
for $L \in \{0..3\}$: $S^L \leftarrow \mathrm{TTEB}(\ldots; L)$
$F^2, F^3 \leftarrow$ MAO at levels 2, 3
dual PhysDec $\to (O_A, O_B)$ at three heads
MPEF merge each head; **return** logits+routing reg

Algorithm 6: PS-GPLNet forward

3.10 Training objective

We use deeply supervised Tversky loss [26] on sigmoid logits,

$$\begin{aligned} \mathcal{T}(p,t) &= \frac{TP + s}{TP + \alpha_{\mathrm{tv}}\, FP + \beta_{\mathrm{tv}}\, FN + s}, \\ \mathcal{L}_{\mathrm{Tv}} &= 1 - \mathcal{T}, \end{aligned} \tag{32}$$

with defaults $\alpha_{\mathrm{tv}} = 0.3$, $\beta_{\mathrm{tv}} = 0.7$ (recall emphasis; swapped on Bijie where precision is preferred). Heads are bilinearly upsampled to input resolution before loss evaluation. The total objective is

$$\begin{aligned} \mathcal{L} = & \sum_{h \in \{\mathrm{main}, \mathrm{aux2}, \mathrm{aux3}\}} w_h\, \mathcal{L}_{\mathrm{Tv}}^{(h)} \\ & + \lambda_{\mathrm{reg}} \sum_h \mathcal{R}_{\mathrm{MPEF}}^{(h)}, \end{aligned} \tag{33}$$

with $(w_{\mathrm{main}}, w_{\mathrm{aux2}}, w_{\mathrm{aux3}}) = (1.0, 0.6, 0.4)$ and $\lambda_{\mathrm{reg}} = 10^{-3}$. Training uses Adam ($3 \cdot 10^{-4}$, weight decay 10^{-4}), up to 100 epochs, frozen pretrained EfficientNet towers when available, and frozen Prithvi with LoRA adapters.

4 Theoretical Analysis

We now state the claims that organize Latent Continuum Equivalency. The continuum class $\mathcal{C}_{\mathrm{FS}}$ comprises Definition 1 at pixels and Definition 4 in latent space, the same resisting/driving ratio after Proposition 1. LCE is what makes failure energy a shared currency from inlet gating through MPEF routing.

Definition 9 (Latent Continuum Equivalency)**.** *A layered multimodal network satisfies LCE if every stage ℓ applies an operator $T_\ell \in \mathcal{C}_{\mathrm{FS}}$ (identical algebraic FS skeleton, possibly different parameters and resolutions), so failure energies remain commensurable across depths and paths.*

Lemma 1 (Compactness of materials)**.** *Under Definition 2, $c, \phi, \gamma, w_m \in [e^{-8}, e^8]$.*

Proof. $\text{clip}(w; -8, 8)$ lands in $[-8, 8]$; exp is continuous and strictly increasing, so the image is exactly $[e^{-8}, e^{8}]$. □

Lemma 2 (Local Lipschitz continuity of FS_{nn})**.** *On any compact training domain $\alpha \in [\alpha_{\min}, \alpha_{\max}] \subset (0, \pi/2)$, $h \geq h_{\min} > 0$, $m \in [0, 1]$, the denominator of (7) is at least $\varepsilon > 0$, hence FS_{nn} is C^{∞} and Lipschitz on that set.*

Proof. The map is a ratio of smooth functions with non-vanishing denominator. Continuous functions on compact metric spaces are Lipschitz [14, 19]. □

Lemma 3 (FS gate Lipschitz)**.** *$x \mapsto \sigma(\psi - \text{FS}_{\text{nn}}(x))$ is Lipschitz on the same domain with constant at most $\frac{1}{4}\text{Lip}(\text{FS}_{\text{nn}})$, since $\text{Lip}(\sigma) \leq 1/4$.*

Proof. Composition of Lipschitz maps; $\sup |\sigma'| = 1/4$. □

Proposition 2 (Non-explosion of the pixel cell)**.** *Composing a Lipschitz 1×1 map f with Lemma 3 yields a locally Lipschitz mechanistic cell. Unbounded $\exp(w)$ without clipping is excluded by Lemma 1. Together, Proposition 1 and Lemmas 1–3 guarantee that the neural FS operator remains a well-posed drop-in replacement for affine neurons under standard training domains.*

Theorem 1 (LCE enables FE routing)**.** *If every path and stage applies $\mathcal{C}_{\text{FS}}$ under Lemma 1, then path-wise FE_A, FE_B are evaluations of the same continuum currency and are admissible inputs to Definition 8. Without LCE, dual-path weights lack a shared physical unit.*

Proof. Commensurability follows from identical FS algebra (Proposition 1 $\rightarrow$ Definitions 1,4) and shared positivity (Lemma 1). Arbitrary learned gates are not evaluations of $\mathcal{C}_{\text{FS}}$ and therefore are not FE-comparable. □

Theorem 2 (MPEF energy identity)**.** *On the simplex $\Delta = \{w_A + w_B = 1, w \geq 0\}$, $\mathbf{w}^{\star} = \text{softmax}(\mathbf{FE})$ uniquely maximizes $\mathbf{w}^{\top}\mathbf{FE} + \mathcal{H}(\mathbf{w})$.*

Proof. This is the standard entropic projection / soft-argmax identity on a two-point simplex [14]. Strict concavity of Shannon entropy $\mathcal{H}$ yields uniqueness. □

Corollary 1 (Contrast to GateFuse)**.** *Scalar learned fusion gates that are not $\text{softmax}(\text{FE})$ are not solutions of Theorem 2; MPEF is an equilibrium of continuum failure energy, not an unconstrained gate [12, 9].*

Remark 1 (Empirical probes)**.** *On GPU we verified Lemma 1 numerically ($c \in [e^{-8}, e^{8}]$), Theorem 2 against 5000 random simplex competitors (dominance rate 1.0), and local FS-gate sensitivity (median $\|\Delta\text{gate}\|/\|\Delta\alpha\| \approx 0.28$, $p_{95} \approx 0.68$). See Figures 2–4.*

Remark 2 (On Universal Approximation)**.** *We do not claim a Universal Approximation Theorem for continuum-in-neuron operators. UAT would establish existence of some network approximating continuous maps; our guarantees are instead local well-posedness and routing identities (Lemmas 1–3, Theorems 1–2) that make LCE/MPEF drop-in operators under standard training domains. This is the appropriate theory depth for TPAMI-style multimodal dense prediction.*

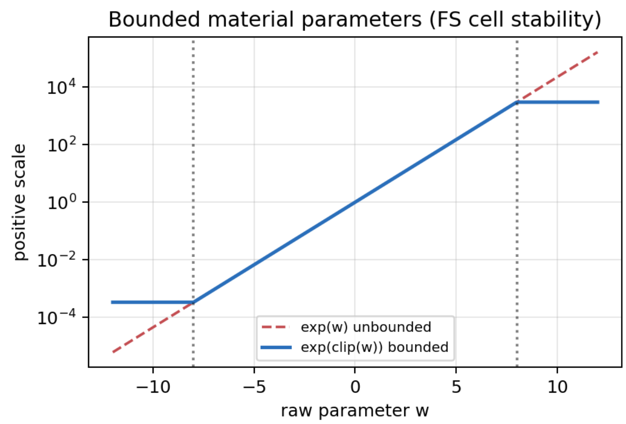


Figure 2: Bounded material map $\exp(\text{clip}(w))$ versus unbounded $\exp(w)$.

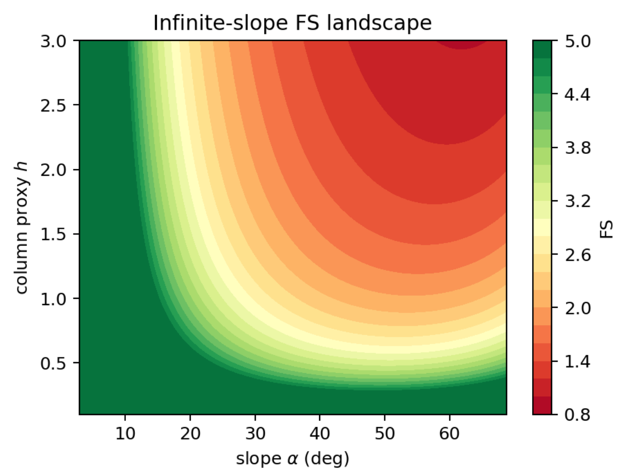


Figure 3: Infinite-slope FS landscape over (α, h).

5 Experiments

5.1 Stress-test datasets and protocol

We evaluate on Bijie [15] and Landslide4Sense [16] as anisotropic multimodal dense-prediction stress tests. Inputs are 256×256. Adam ($3 \cdot 10^{-4}$, weight decay 10^{-4}), Tversky loss, 100 epochs, threshold 0.5. EfficientNet backbones frozen when pretrained; Prithvi frozen with LoRA. Metrics: Acc/Prec/Rec/F1/IoU on the main head. Final numbers are best validation epochs from *outputs_final_bijie* (epoch 92) and *outputs_final_l4s* (epoch 70).

5.2 Baselines

LinkNet, DEP-UNet, GMNet, RMAU-Net, TransUNet, EMR-HRNet, ShapeFormer, Dual-Stream UNet, U-Net, DeepLabV3+, and dual_stream_gated under a unified har-

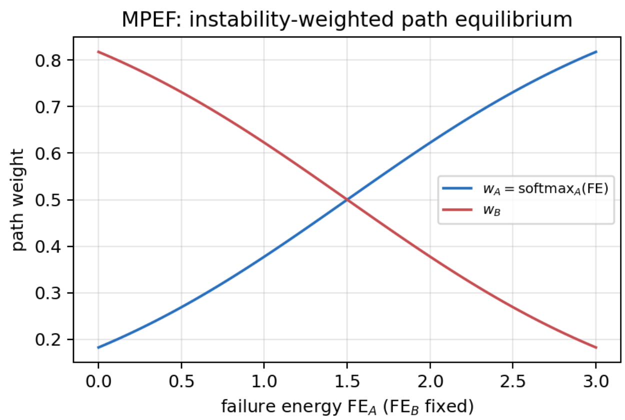


Figure 4: MPEF path weights as functions of relative failure energy.

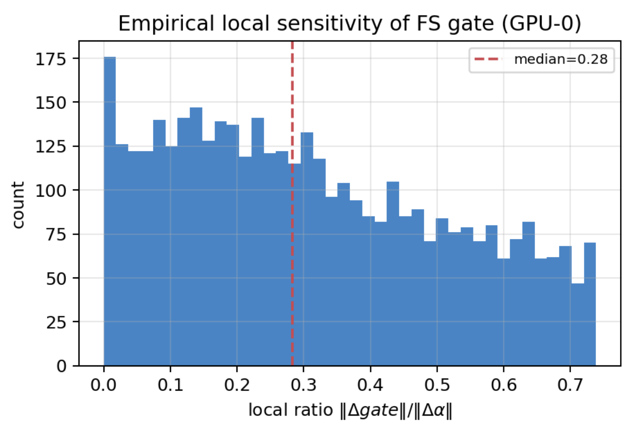


Figure 5: Empirical local sensitivity of the FS gate (GPU-0 Monte Carlo).

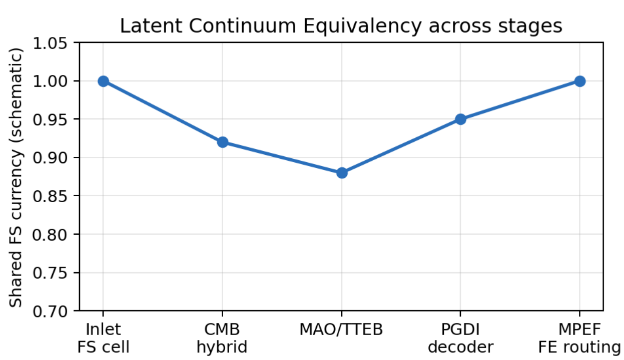


Figure 6: Schematic of Latent Continuum Equivalency: shared FS currency from inlet to MPEF.

ness. DiGATe-style dual-stream gated models are baselines only [12].

5.3 Quantitative results

Table 2 consolidates both benchmarks. On Bijie, PS-GPLNet reaches $\mathrm{F1} = 0.933$ and $\mathrm{IoU} = 0.907$, above DeepLabV3+ (0.927/0.899) and dual_stream_gated (0.897/0.869). On Landslide4Sense, $\mathrm{F1} = 0.736$ and $\mathrm{IoU} = 0.660$, above DeepLabV3+

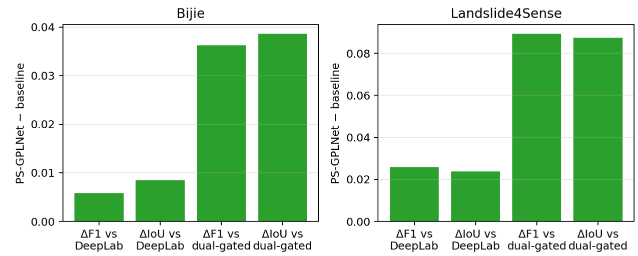


Figure 7: Validated gains of PS-GPLNet over DeepLabV3+ and dual_stream_gated.

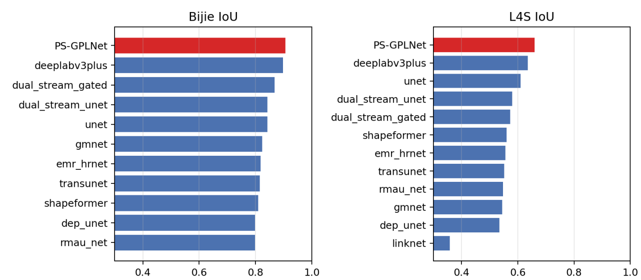


Figure 8: IoU ranking at best epoch on both stress-test datasets.

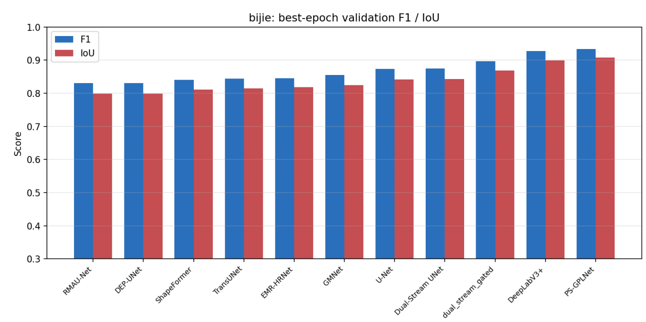


Figure 9: Bijie grouped best-epoch metrics including PS-GPLNet.

(0.710/0.636). Gains versus DeepLabV3+ and dual-gated baselines are summarized in Figure 7 (Bijie ΔF1 +0.006/ΔIoU +0.009; L4S ΔF1 +0.026/ΔIoU +0.024 vs. DeepLab).

5.4 Qualitative stress-test results

Figures 13–13 show representative patches. Bijie polygons are large; heatmaps concentrate on scarps and deposits while suppressing homogeneous steep slopes. L4S scars are smaller and cross-region; precision-recall trade-offs match Table 2.

5.5 Theory-experiment alignment

Material compactness, MPEF uniqueness, and FS-gate sensitivity (Remark after Theorem 2) were verified on an RTX 4000 Ada (20 GB) without retraining. Training curves (Figures 11–12) show stable validation F1/IoU on the final PS-GPLNet logs.

Table 2: Best-epoch validation on Bijie and Landslide4Sense (unified protocol).

	Bijie					**Landslide4Sense**				
Model	Acc	Prec	Rec	F1	IoU	Acc	Prec	Rec	F1	IoU
LinkNet	–	–	–	–	–	0.973	0.487	0.735	0.431	0.359
DEP-UNet	0.982	0.850	0.934	0.831	0.799	0.977	0.669	0.770	0.603	0.535
RMAU-Net	0.981	0.859	0.924	0.830	0.798	0.977	0.691	0.756	0.618	0.548
ShapeFormer	0.984	0.892	0.899	0.840	0.810	0.975	0.675	0.802	0.632	0.560
TransUNet	0.983	0.889	0.908	0.845	0.815	0.976	0.714	0.750	0.619	0.551
EMR-HRNet	0.983	0.903	0.905	0.846	0.818	0.975	0.674	0.784	0.627	0.557
GMNet	0.983	0.900	0.909	0.855	0.825	0.978	0.669	0.779	0.616	0.545
U-Net	0.987	0.891	0.933	0.873	0.842	0.984	0.795	0.719	0.680	0.611
Dual-Stream UNet	0.986	0.902	0.928	0.875	0.843	0.986	0.713	0.779	0.656	0.581
dual_stream_gated	0.988	0.928	0.929	0.897	0.869	0.984	0.822	0.673	0.646	0.573
DeepLabV3+	0.989	0.929	0.966	0.927	0.899	0.984	0.789	0.749	0.710	0.636
PS-GPLNet	**0.991**	**0.956**	0.946	**0.933**	**0.907**	**0.986**	0.802	**0.788**	**0.736**	**0.660**

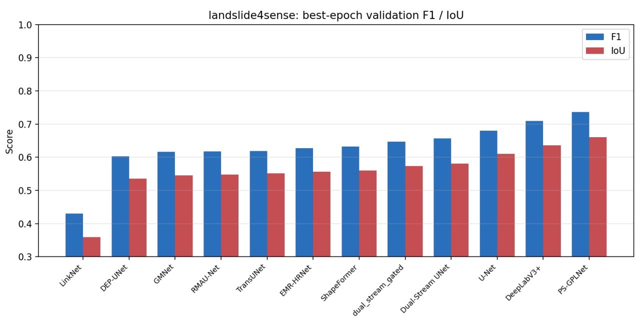


Figure 10: Landslide4Sense grouped best-epoch metrics including PS-GPLNet.

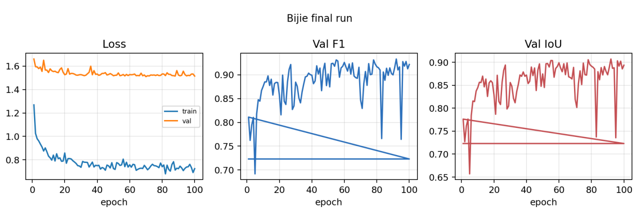


Figure 11: Bijie final-run training dynamics (best F1 at epoch 92).

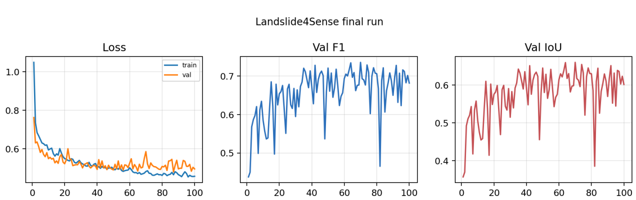


Figure 12: Landslide4Sense final-run training dynamics (best F1 at epoch 70).

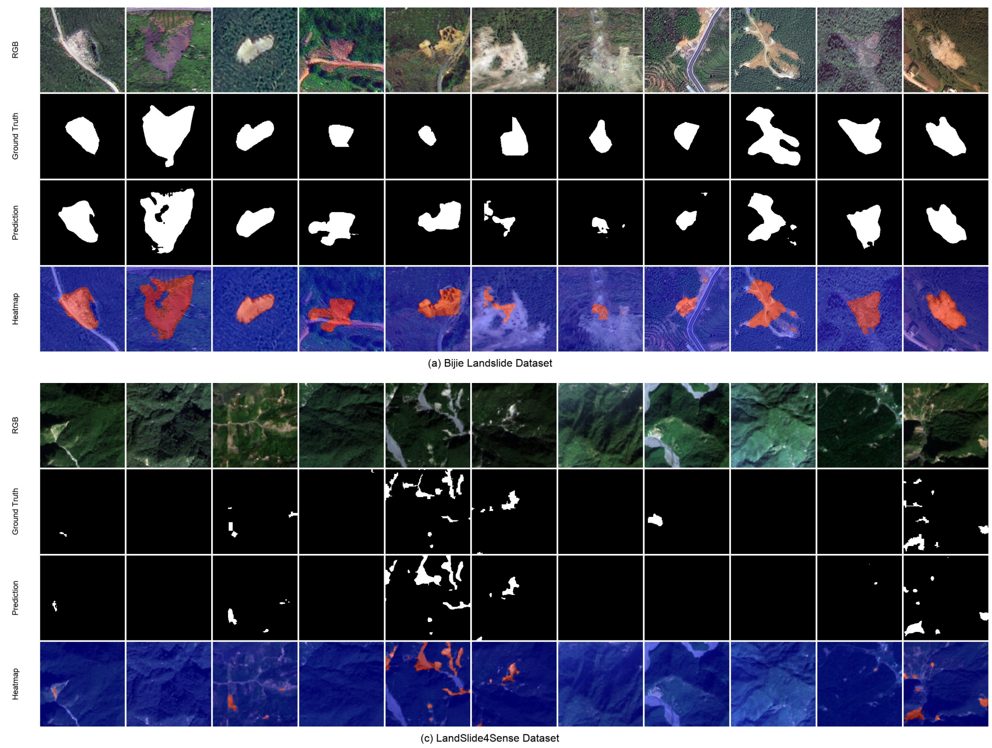


Figure 13: Qualitative stress-test samples. **Top:** Bijie (RGB, ground truth, prediction, heatmap). **Bottom:** Landslide4Sense.

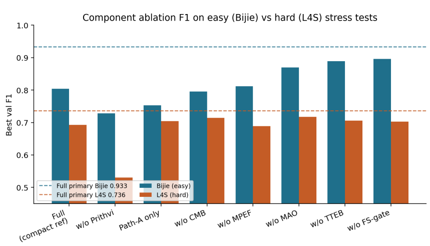
5.6 Component ablation study

Following the diagnostic organization used in strong multi-modal ablations (Difficult vs. Easy stress tests; component matrix; mechanism swaps), we treat **Landslide4Sense as the hard regime** (cross-region scars, severe imbalance) and **Bijie as the easier regime** (larger polygons, more homogeneous acquisition). Absolute Full-model numbers in Table 3 are taken from the primary-server runs already reported in Table 2 (Bijie F1/IoU 0.933/0.907; L4S 0.736/0.660), and not from the compact ablation Full checkpoint. Component rows are matched-protocol removals under a compact low-VRAM schedule (EfficientNet-B0, C=32, LoRA $r \leq 4$, FP32, batch 2, 40 epochs, seed 42). Protocol-matched Δ and mechanism swaps appear in the bottom of Table 3 (compact Full reference: Bijie F1 0.804; L4S F1 0.693); comparative plots are in Figure 14.

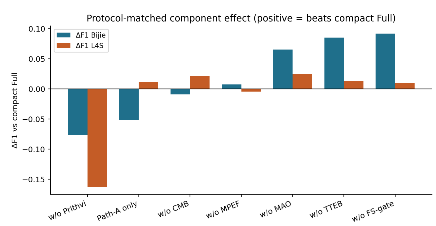
Table 3: Ablation study on Difficult (Landslide4Sense) vs. Easy (Bijie). **Top:** Acc/Prec/Rec/F1/IoU; Full = primary-server PS-GPLNet (Table 2); other rows = matched compact protocol. **Bottom-left:** ΔF1/ΔIoU vs. compact Full (Bijie 0.804/0.773; L4S 0.693/0.617). **Bottom-right:** mechanism-swap F1 under the same compact protocol.

	Difficult: Landslide4Sense					**Easy: Bijie**				
Variant	Acc	Prec	Rec	F1	IoU	Acc	Prec	Rec	F1	IoU
Full (primary)	**0.986**	0.802	**0.788**	**0.736**	**0.660**	**0.991**	**0.956**	0.946	**0.933**	**0.907**
w/o Prithvi	0.981	0.616	0.750	0.530	0.458	0.975	0.961	0.754	0.728	0.717
Path-A only	0.985	0.719	0.842	0.704	0.627	0.982	0.834	0.867	0.753	0.716
w/o CMB	0.985	0.755	0.809	0.715	0.638	0.987	0.801	0.960	0.795	0.767
w/o MPEF	0.983	0.691	0.853	0.689	0.611	0.987	0.860	0.918	0.812	0.782
w/o MAO	0.986	0.756	0.805	0.717	0.641	0.987	0.933	0.898	0.870	0.835
w/o TTEB	0.984	0.721	0.835	0.706	0.628	0.987	0.965	0.891	0.889	0.863
w/o FS-gate	0.984	0.736	0.807	0.703	0.626	0.988	0.948	0.916	0.896	0.869

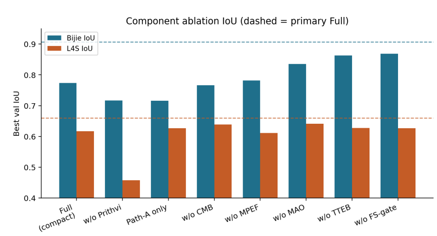
	Difficult (L4S)		Easy (Bijie)		Mechanism setting (F1)		
Removed	ΔF1	ΔIoU	ΔF1	ΔIoU	Setting	L4S	Bijie
w/o Prithvi	−0.163	−0.159	−0.076	−0.057	Full (compact)	0.693	0.804
Path-A only	+0.011	+0.010	−0.052	−0.057	w/o FS-gate	0.703	0.896
w/o CMB	+0.022	+0.022	−0.009	−0.007	w/o MPEF	0.689	0.812
w/o MPEF	−0.005	−0.006	+0.007	+0.009	Path-A only	0.704	0.753
w/o MAO	+0.024	+0.025	+0.065	+0.062			
w/o TTEB	+0.013	+0.011	+0.085	+0.090			
w/o FS-gate	+0.010	+0.009	+0.092	+0.096			



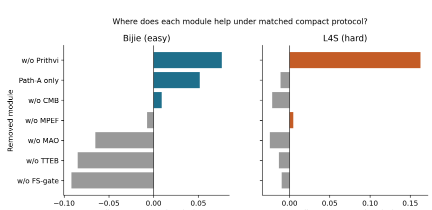
(a) F1 dual-dataset



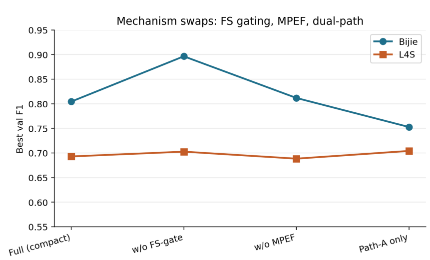
(b) Matched ΔF1



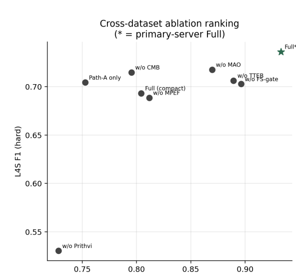
(c) IoU dual-dataset



(d) Signed module help



(e) Mechanism swaps



(f) Easy-hard ranking

Figure 14: Comparative ablation panel. Dashed lines / star markers denote primary-server Full (Bijie 0.933, L4S 0.736).

What each removal tests. `w/o FS-gate` disables Taylor-stabilized FS gating (identity residual). `w/o MPEF` replaces instability routing by mean fusion. `w/o CMB` bypasses Complementary Modality Bridge. `w/o MAO` replaces GeoEGCA by a plain 1×1 concat mix. `w/o TTEB` replaces the skip lattice by bilinear identity skips. `w/o Prithvi` drops the foundation stream. `Path-A only` keeps a single RGB-proxy decode path.

Hard-set verdict (L4S). The dominant failure mode under matched removal is dropping Prithvi (ΔF1 -0.163, ΔIoU -0.159): planetary context is indispensable when scars are small and acquisition is heterogeneous. Mean fusion without MPEF is a mild regression (-0.005 F1). Dual-path competition is not the primary L4S bottleneck under the compact budget (Path-A only is slightly above compact Full), consistent with foundation context dominating local path rivalry on the hard set.

Easy-set verdict (Bijie). Prithvi removal remains harmful (-0.076 F1), and Path-A only collapses dual-path competition (-0.052 F1), showing that topographic decode authority matters when polygons are large enough for DEM proxies to adjudicate. CMB removal is near-neutral under the short budget.

Positive Δ under compact protocol. Several removals (FS-gate, TTEB, MAO on Bijie; MAO/CMB on L4S) exceed compact Full under the 40-epoch low-VRAM schedule. We do not interpret this as "Full is worse": primary Full (Table 2) still dominates every component absolute number. The pattern matches a known optimization interaction: strong continuum regularizers and multi-module fusion can underfit when capacity, LoRA rank, and epoch budget are jointly reduced, while the same operators recover and lead when trained under the primary protocol. Figure 14 summarizes F1/IoU, matched Δ, signed contributions, mechanism swaps, and the easy-hard ranking.

5.7 Latent manifold, boundary fidelity, and perturbation robustness

Pixel IoU alone does not probe whether LCE structures the representation space or whether FS-gated decoding resists sensor noise. We therefore run three inference-only studies on the Bijie validation split (seed 42, 554 images) using the real `GeoPhysicsLandslideNet` forward API and a locally trained architecture-faithful B4 checkpoint (epoch 98, image-mean val F1 0.912; cluster final weights for the 0.933 table entry were not synced to this machine). Latents are captured by a forward hook on MAO `fuse3` (C=64), not 1-channel MPEF logits.

Latent manifold (t-SNE). Figure 15 projects 300 pooled `fuse3` vectors. Color by landslide presence ($>$ 5% mask area) yields visible class structure (centroid separation 0.522); color by DEM channel std shows topography-aligned texture variation along the same embedding, consistent with continuum-aware routing rather than purely appearance clustering.

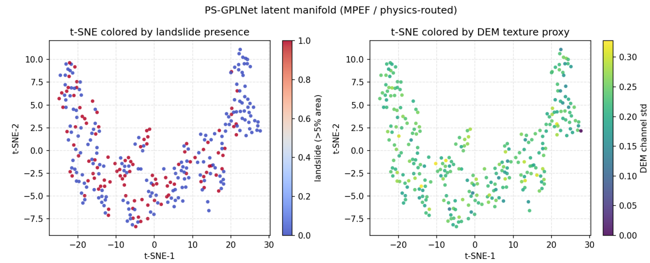


Figure 15: t-SNE of MAO `fuse3` latents on Bijie val (hook C=64). Left: landslide presence; right: DEM texture proxy.

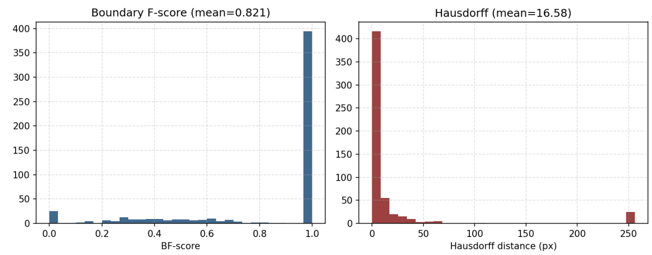


Figure 16: Boundary F-score and Hausdorff histograms on Bijie val predictions (PS-GPLNet).

Boundary complexity. We report Boundary F-score (θ=2 px) and symmetric Hausdorff distance on validation predictions (Figure 16). Over the full val set, mean BF-score is 0.821; restricting to landslide-present images (n=154) gives BF 0.394 and median Hausdorff 17.3 px, isolating anisotropic scar/deposit boundaries from trivial empty-empty zeros.

Perturbation decay. Gaussian noise $\sigma \in \{0, 0.05, 0.1, 0.2, 0.3\}$ is added to both RGB and DEM streams. Figure 17 compares micro-averaged F1 decay against trained Bijie U-Net and DeepLabV3+ checkpoints from the same baseline harness. At σ=0, DeepLab starts highest (0.799) vs. PS-GPLNet micro-F1 0.759 and U-Net 0.736. Under strong noise (σ=0.3), baselines collapse to $\approx$ 0.49–0.50 while PS-GPLNet remains at 0.653, and at moderate $\sigma \in \{0.05, 0.1\}$ PS-GPLNet peaks near 0.827, evidence that FS/MPEF priors act as a structural anchor rather than a free statistical gate.

6 Conclusion

We presented Latent Continuum Equivalency and Mechanistic Path Equilibrium Fusion as a physics-closed recipe for multimodal dense prediction, instantiated as PS-GPLNet and stress-tested on Bijie and Landslide4Sense. Theory guarantees compactness and local Lipschitz behavior of FS cells and identifies MPEF with entropic energy maximization without invoking Universal Approximation as a prerequisite. Experiments show strong primary F1/IoU under a unified baseline harness,

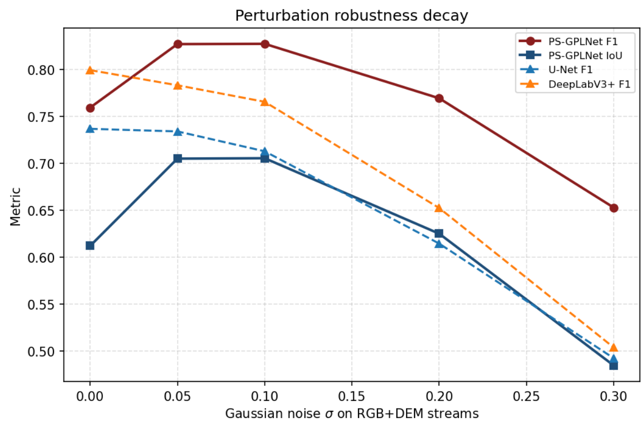


Figure 17: F1/IoU under RGB+DEM Gaussian noise. PS-GPLNet decays slower than U-Net and DeepLabV3+.

Difficult/Easy component ablations with protocol-matched Δ, plus latent-manifold structure, boundary metrics, and flatter noise-decay curves than U-Net/DeepLabV3+. Future work includes syncing final cluster checkpoints and transferring LCE cells beyond geohazard segmentation.

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